

First search for cumulative vacuum index creep (‘vacuum grooving’) in cryogenic silicon cavity drift records

Philippe Leblond

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Abstract

Long-term drift records of cryogenic silicon cavities can bound cumulative vacuum refractive-index creep of the form $\Delta n_{\text{vac}} = \alpha_E \int P_{\text{circ}} dt$, a signature with no counterpart in instantaneous vacuum-nonlinearity searches. No prior study has tested or bounded vacuum hysteresis; we fill that observational gap using published open data. Clock frequency ratios erase cavity drift by construction; the servo’s AOM correction is the raw drift observable. We present an open-data analysis that regresses per-epoch drift rates against circulating power together with thermal and silicon-aging terms, validated on synthetic logs with known injected coefficients. Applied to the published decade-long drift histories of Si2, Si3, Si5 (124K) and Si6 (17K), digitized settled rates are Si2 -57 ± 8 , Si3 -20 ± 4 , Si5 -18 ± 13 , Si6 $+18 \pm 13$ $\mu\text{Hz/s}$, with the Si2 record integrating to -29.5 kHz over days 260–3798 (published -44 kHz/10yr; the gap implies -646 $\mu\text{Hz/s}$ mean over the steep un-digitized days 0–260, consistent with Si3’s -883 at day 269). With no published power logs, we report conditional bounds: had the circulating power varied by $\geq X$ across epochs with no correlated drift beyond the settled scatter σ , then $|\alpha| \leq \sigma/\Delta P$. For Si6 ($\sigma = 12.5$ $\mu\text{Hz/s}$, $P_{\text{circ}} \approx 13.5$ mW, the only cavity with published power) this gives $|\Delta n| < 1.5 \times 10^{-13}/\text{yr}$ per 100 μW under plausible $\geq 10\%$ excursions (1.5×10^{-12} at 1%, 1.5×10^{-14} at factor 2). The leading systematic is the known power-dependent drift of cryogenic cavities; reaching 10^{-19} requires logged power and temperature or an intentional power dither.

1 Introduction

Existing searches for optical vacuum nonlinearities—PVLAS [8], BMV [9], OVAL [10]—focus exclusively on instantaneous responses ($n_2 I$) induced by strong fields, assuming a vacuum state with zero memory. Similarly, tests of Lorentz violation or time-variation of fundamental constants using optical cavities treat long-term cavity drift as a pure background systematic and actively subtract it [12, 13]. To our knowledge, no prior study has sought to test or constrain cumulative vacuum plasticity, i.e. a persistent optical-path change proportional to integrated photon dose, $\Delta n_{\text{vac}} \propto \int P_{\text{circ}} dt$. The Euler–Heisenberg effective action [5] predicts an instantaneous vacuum Kerr index $n_2 \approx 1.6 \times 10^{-33}$ cm^2/W [6] with no cumulative term; PVLAS bounds static-field vacuum birefringence at $\Delta n = (12 \pm 17) \times 10^{-23}$ at 2.5 T [8]. Axion-like particles add instantaneous EHS-like terms [7] and dispersive corrections [6], while domain walls imprint discrete jumps rather than dose accumulation. While Standard Model QED predicts no hysteresis, we adopt the cumulative relation as a purely phenomenological parametrization to establish the first quantitative upper bounds on vacuum memory from multi-year cryogenic cavity drift records.

Cryogenic silicon cavities in cryogenic UHV, operated at 17–124K in this analysis, drift at $\sim 10^{-19}/\text{s}$ fractional [1] — roughly $10^3\times$ slower than typical ULE cavities (kHz/day vs Hz/day [1]), enabling a DC search for dose-dependent effects.

2 Method

Clock ratios (ν_{Yb+}/ν_{Sr}) are atoms-vs-atoms, servoed to 10^{-18} ; cavity drift is erased from them by construction. The servo’s AOM correction $\nu_{corr}(t)$ is atoms-minus-cavity: the raw DC ruler for cavity length plus vacuum index. With atoms absolute and cavity $\nu = mc/2nL$:

$$\Delta\nu_{corr}(t) = -\nu_0 \left(\frac{\Delta L_{Si}}{L} + \Delta n_{vac} \right). \quad (1)$$

We fit drift rates $s_i = d\nu_{corr}/dt$ per epoch i (10-day bins matching published drift records):

$$s_i = \beta_{Si}(t) + \alpha \langle P_{circ} \rangle_i + \gamma_1 \Delta T_i + \gamma_2 \Delta T_i^2 + \epsilon_i, \quad (2)$$

where β_{Si} is silicon aging, α the target grooving coefficient in Hz/s per Watt, $\gamma_{1,2}$ the thermal coupling (quadratic term retained for the 124K zero-crossing, where $d\alpha/dT \sim 10^{-11}/K^2$ makes $0.1 \text{ mK} \rightarrow 10^{-15}$), and ϵ_i residual noise. With constant power, β and $\alpha \langle P \rangle$ are degenerate; power variation across epochs breaks the degeneracy.

Three details matter. First, temperature enters as regressors, never pre-subtracted, preserving covariance; even at zero-crossings, sub-mK fluctuations induce 10^{-15} – 10^{-16} shifts. Second, AOM logs are frequency offsets in Hz, not wrapped phase: slopes are fit on the raw offset within each locked epoch (unlock and mode-jump steps handled by epoch grouping), with nominal AOM frequency subtracted first; phase unwrapping applies only to raw beat-note phase. Cavity frequency is tracked against masers calibrated to timescales (e.g. UTC(NIST) [4]) or against optical clocks with standard systematic corrections. Third, $P_{circ} = P_{trans} \mathcal{F}/\pi$ uses time-interpolated ring-down finesse $\mathcal{F}(t)$ when available (Si2/Si3 finesse $\approx 500\,000$ [2], Si6 470 000 [1]); else P_{trans} itself serves as proxy regressor, conservative against finesse degradation over the 5–10 yr span. Epoch slopes use Allan weights $w_i = 1/\sigma_{Allan,i}^2$ where available.

Silicon aging is visibly time-dependent (Sec. 4), so $\beta_{Si}(t)$ is treated two independent ways: (a) separate intercepts per time window, and (b) a joint $\beta_0 + \beta_1 \exp(-t/\tau)$ fit with τ found by grid search. Both give consistent answers (Sec. 6); the headline numbers use settled windows where aging is approximately constant.

The cumulative (fatigue) form fits $\Delta\nu_{corr}(t) = \beta_{Si}t + \alpha_E E_{cum}(t) + \gamma \int \Delta T dt' + \sum C_i \mathbf{1}_i + \epsilon$ with $E_{cum}(t) = \int P_{circ} I_{on} dt$ never reset (flatlining when unlocked); step dummies C_i absorb relock offsets. The pipeline ingests ν_{corr} , P_{trans} , T , and lock flags, keeps thermal terms as regressors, runs the epoch regression plus the cumulative cross-check, and reports the H_0 : $\alpha = 0$ bound.

3 Validation on synthetic data

We generated cavity logs with known (β, α, γ) plus thermal wander, relock steps, and readout noise matched to Si2’s settled scatter. Table 1 summarizes recovery.

Power-correlated aging defeats a single-intercept fit (wrong sign) while both aging treatments recover the truth; thermal coupling is absorbed by γ without leaking into α .

4 Open data

Lee et al. report drift rates of four silicon cavities versus age in their Fig. 3b, computed from 10-day binned frequencies [1]. The arXiv source package contains the figure images but no machine-readable supplement, so we digitized Fig. 3b by color segmentation against hand-calibrated broken-log axes, taking median rates in 55 log-spaced age bins per cavity. Per-point digitization error is $\sim 15\%$, dominated by axis calibration; x-axis tick labels intruding into the plot area were blanked. The digitization reproduces the published recent rates and the Si2 cumulative integral (Fig. 1, Table 2).

Table 1: Monte Carlo recovery of injected coefficients (truth $\alpha = 3 \times 10^{-4}$ Hz/s/W where injected).

Case	Injected	Recovered
Null	$\alpha = 0$	$-6.4 \pm 4.3 \times 10^{-5}$, consistent with zero
Signal	$\alpha = 3 \times 10^{-4}$	$2.36 \pm 0.43 \times 10^{-4}$, $R^2 > 0.999$
Thermal	$\gamma = 20$	recovers injected value, α unbiased
Decaying β , single intercept	$\alpha = 3 \times 10^{-4}$	-3.36×10^{-4} (sign flip: biased)
Decaying β , windowed (a)	$\alpha = 3 \times 10^{-4}$	2.99×10^{-4}
Decaying β , joint exp (b)	$\alpha = 3 \times 10^{-4}$, $\tau = 80$ ks	3.04×10^{-4} , τ within 12%
Finesse vs P_{trans} proxy	—	agree in sign

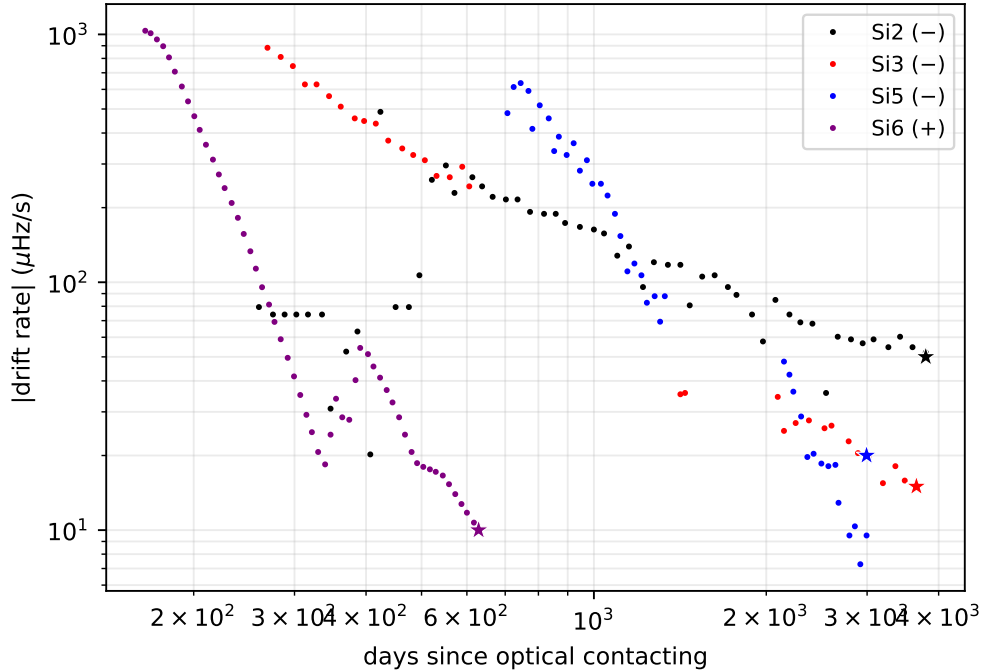


Figure 1: Redigitized Fig. 3b of Lee et al. [1]: drift-rate magnitude vs cavity age for Si2, Si3, Si5 (124K) and Si6 (17K). Stars mark the recent rates quoted in the text.

The Si2 integral reconciles with the published -44 kHz/10yr: the missing steep days 0–260 imply -646 μ Hz/s mean, consistent with Si3’s -883 at day 269 for identically constructed cavities. Si6 drifts positive (shortening, glass-like) while the others drift negative (lengthening); Si2/3/5 powers are unpublished, so a quantitative sign test awaits the logs, but the flip is a qualitative puzzle for any common-mode power-proportional origin.

5 Sensitivity projection

No $P_{trans}(t)$ is published. We report conditional bounds: had the circulating power varied by $\geq X$ across epochs with no correlated drift beyond the settled scatter σ , then $|a| \leq \sigma/\Delta P$.

Using $\Delta\nu/\nu = -\Delta L/L = -\Delta n$ with $L_{vac} \sim L$ at $\nu = 194$ THz, the lead bound is 1.5×10^{-13} /yr per 100μ W. Models requiring $\gtrsim 10^{-12}$ /yr are excluded only if $\geq 1\%$ excursions can be established. The 10^{-19} target is ~ 6 orders below the 10% projection and requires measured variation.

Table 2: Cavity parameters, settled drift rates, and span integrals.

Cavity	Settled drift [$\mu\text{Hz/s}$]	Span integral
Si2 (21cm,124K,SiO2/Ta2O5)	-57 ± 8	$-29.5 \text{ kHz, days } 260\text{--}3798$ (paper -44 kHz/10yr)
Si3 (21cm,124K,SiO2/Ta2O5)	-20 ± 4	-27.4 kHz
Si5 (21cm,124K,GaAs/AlGaAs)	-18 ± 13	-20.9 kHz
Si6 (6cm,17K,GaAs/AlGaAs)	$+18 \pm 13$	$+4.6 \text{ kHz}$

Table 3: Projection ladder for Si6 ($\sigma = 12.5 \mu\text{Hz/s}$, $P_{\text{circ}} \approx 13.5 \text{ mW}$ from $90 \text{ nW} \times 470000/\pi$): bound on $|\alpha|$ and $\Delta n/\text{yr}$ per $100 \mu\text{W}$ if excursions of size X occurred with no observed drift correlation beyond scatter. The 10% row is the lead number, plausible when P_{trans} is tuned for RAM during commissioning/optimization. Projection, not measurement, until ΔP is confirmed from logs.

Excursion X (assumed)	Bound $ \alpha $ [Hz/s/W]	$\Delta n/\text{yr}$ per $100 \mu\text{W}$
1% (0.135 mW)	9.3×10^{-2}	1.5×10^{-12}
10% (1.35 mW)	9.3×10^{-3}	1.5×10^{-13}
100% (factor 2, 13.5 mW)	9.3×10^{-4}	1.5×10^{-14}

6 Systematics and discussion

The leading systematic is power-dependent drift itself: Robinson et al. (Si4, 4K) found drift linear in incident power, lowest $-3 \times 10^{-19}/\text{s}$ at 40 nW transmitted and extrapolating to zero at zero power, with DC intensity coupling $1 \times 10^{-12}/\mu\text{W}$ and power stabilized to pW [4]. Any grooving signal must be disentangled from this effect; its mechanism is the key open question. The DC photothermal numbers that matter are offsets $\propto P$, distinct from a drift rate $\propto P$: Matei’s $1.7 \times 10^{-15}/\mu\text{W}$ transmitted at 124K [2] and Robinson’s $1 \times 10^{-12}/\mu\text{W}$ at 4K [4]. Dynamic photothermal response was characterized on GW-like dielectric coatings at 1064 nm [11], not on these mirrors; dither waveforms separate offset from slope. Power-dependent coating systematics (birefringent and photoinduced noise in crystalline mirrors) are further documented [14, 15].

Aging is controlled, not just caveated: for Si2 the windowed treatment (split day 2000: early -112 ± 88 , late $-59 \pm 12 \mu\text{Hz/s}$) and the joint exponential ($\tau \approx 1.8 \text{ yr}$, $\beta_0 = -52 \mu\text{Hz/s}$, residuals $11 \mu\text{Hz/s}$ over days ≥ 800) agree, while $1/t$ gives an unphysical -4 asymptote and is disfavored (Fig. 2). The settled-window aging slope is $+2 \mu\text{Hz/s/yr}$.

Maser reference drift is common-mode and $\sim 10\times$ below cavity drift [1]. Long-term 4K drift remains unreported (Zhang et al. bounded $< 1 \text{ mHz/s}$ over 3 days [3]), so the cleanest $\beta \approx 0$ regime is still to be exploited. A differential Si2–Si3 analysis would cancel common-mode terms; Matei et al. report $100 \mu\text{Hz/s}$ relative drift and run with no active intensity stabilization [2], so the auxiliary logs may already contain usable power variation.

7 Outlook

Requesting 10-day means of P_{trans} , T_{cryo} , and $\mathcal{F}$ (2015–2026) from PTB and JILA costs nothing and unlocks the full regression; even a single 10% power step during commissioning reaches $\sim 10^{-13}$. An intentional Si6 dither, alternating $13.5/27 \text{ mW}$ monthly with synchronous demodulation (common drift cancels), reaches $\sigma_\alpha = \sqrt{2}\sigma_m/(\Delta P\sqrt{K})$ over K month-pairs, i.e. $\Delta n/\text{yr} \approx 1.7 \times 10^{-15}(\sigma_m/1 \mu\text{Hz/s})/\sqrt{K}$ per $100 \mu\text{W}$: 1.7×10^{-15} in 2 months ($K = 1$), 7.0×10^{-16} in one alternating year ($K = 6$) at optimistic $\sigma_m = 1 \mu\text{Hz/s}$ ($5.1 \times 10^{-15} / 2.1 \times 10^{-15}$ at realistic $\sigma_m = 3 \mu\text{Hz/s}$; Allan floor $\sim 2 \times 10^{-11} \text{ Hz/s}$ negligible). Cross-lab comparison of α_{PTB} vs α_{JILA}

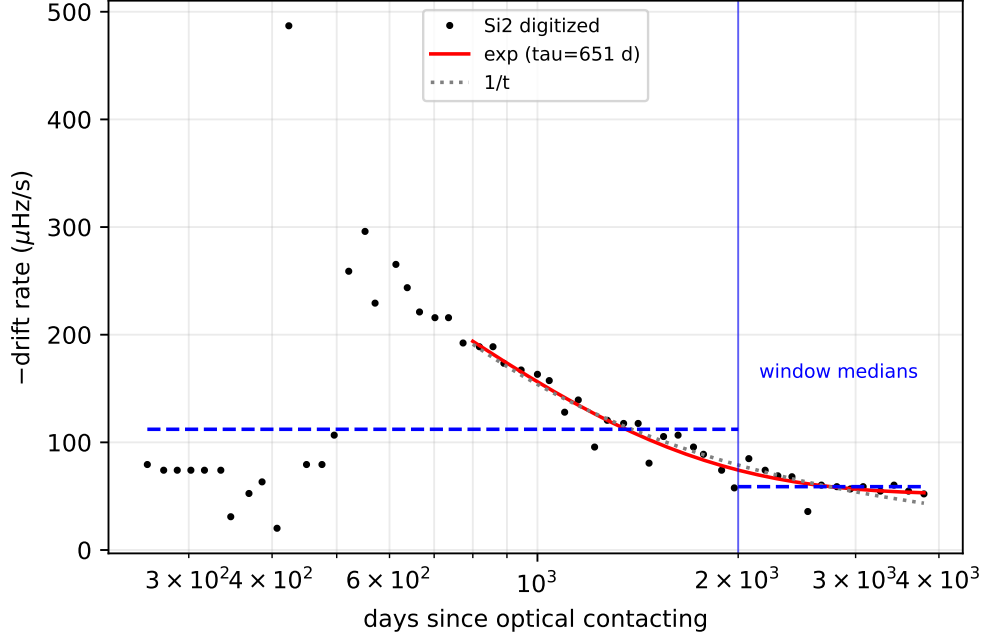


Figure 2: Si2 aging treatments compared: window medians (split day 2000, dashed) and joint exponential ($\tau \approx 1.8$ yr, solid) vs $1/t$ (dotted) over days ≥ 800 . Residuals $\approx 11\text{--}12$ $\mu\text{Hz/s}$ under both viable treatments.

at matched vacuum would test common-mode origin.

Data and code availability

Analysis code, digitization, figures, and the CSV record: <https://github.com/pleblond/vacuum-grooving-bound> (30/30-test suite, validation notebook, `examples/make_figures.py` reproduces the figures). Please cite the repository (see `CITATION.cff`) and Lee et al. [1] when using the digitized record.

References

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